CS & IT ENGINEERING

DIGITAL LOGIC

Combinational Circuit



Lecture No. 06



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01 DEMUX

02 Question Practice

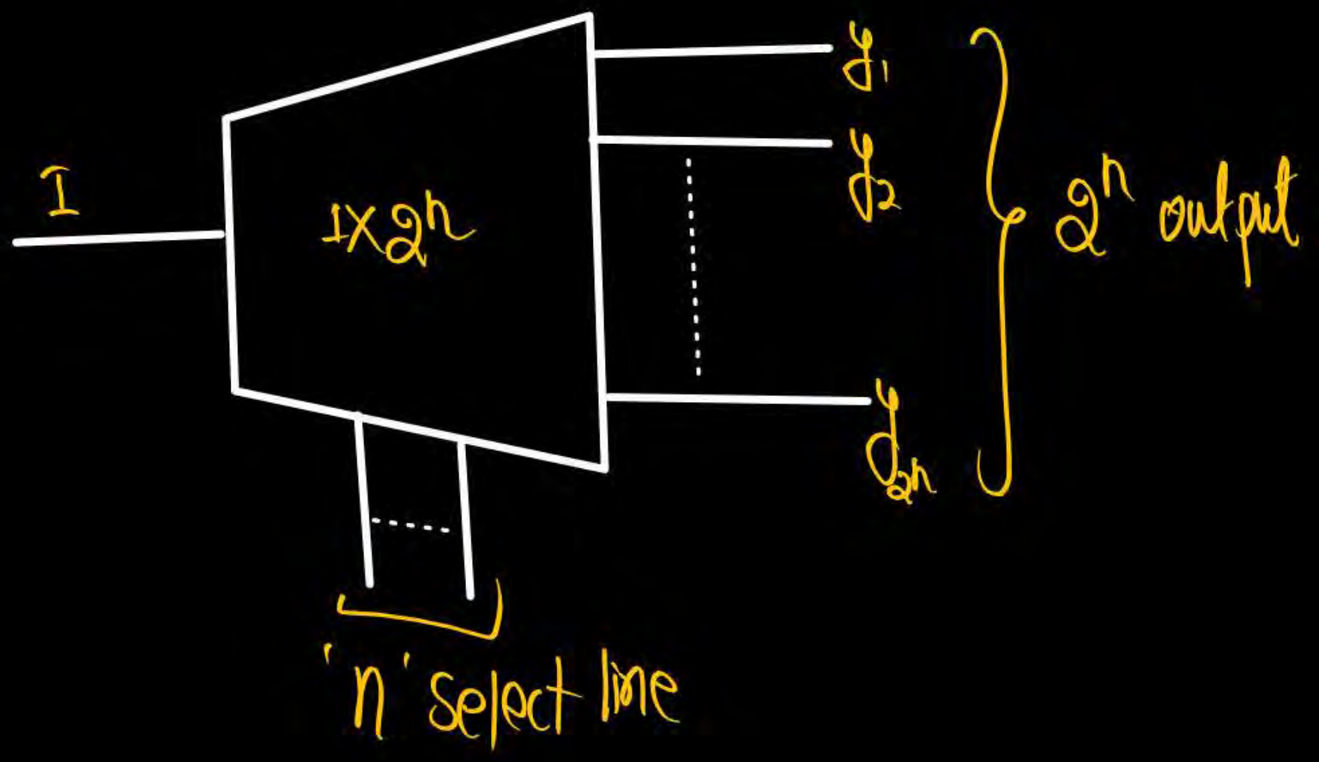
03 ENCODER

03 Discussion

DEMUX

What is DeMUX ?

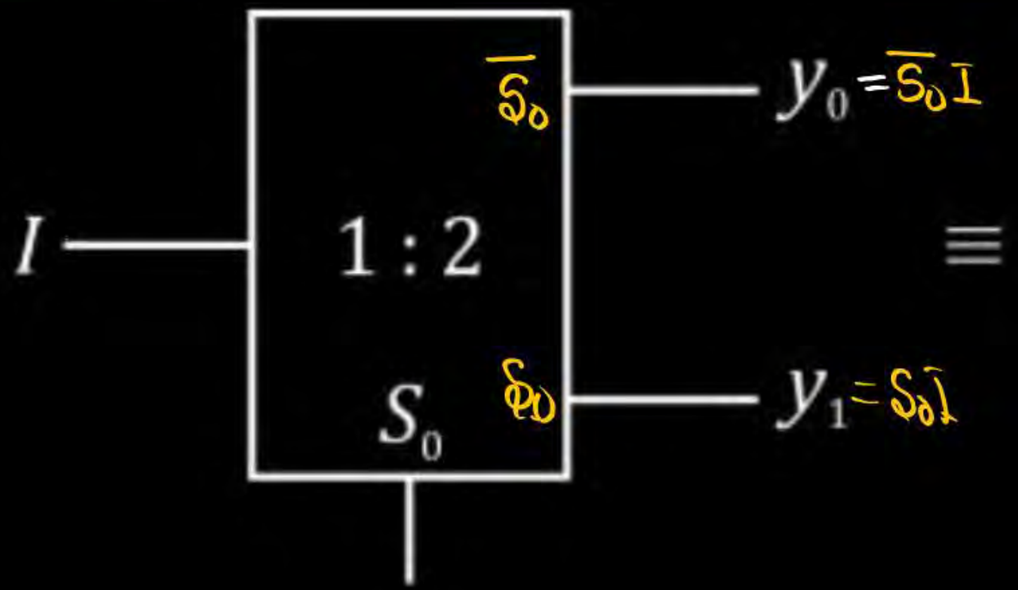
↳ Data Distributor



DEMUX

Design a 1 x 2 DE-MUX?

Step 1:



Step 2:

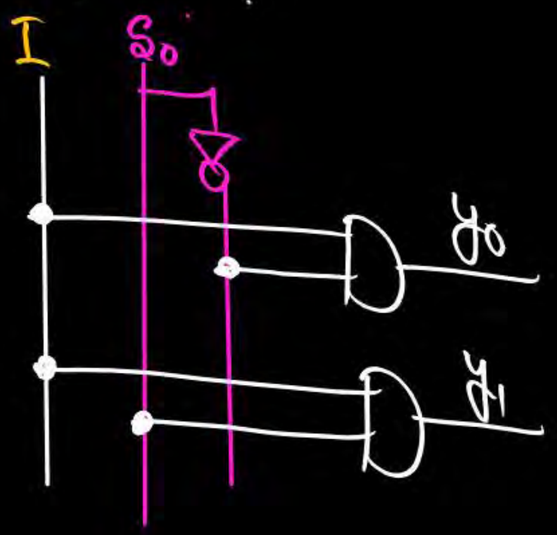
S_0	y_0	y_1
0	I	0
1	0	I

Step 3: $y_0 = \bar{S_0} I$

$y_1 = S_0 I$

Step 4:

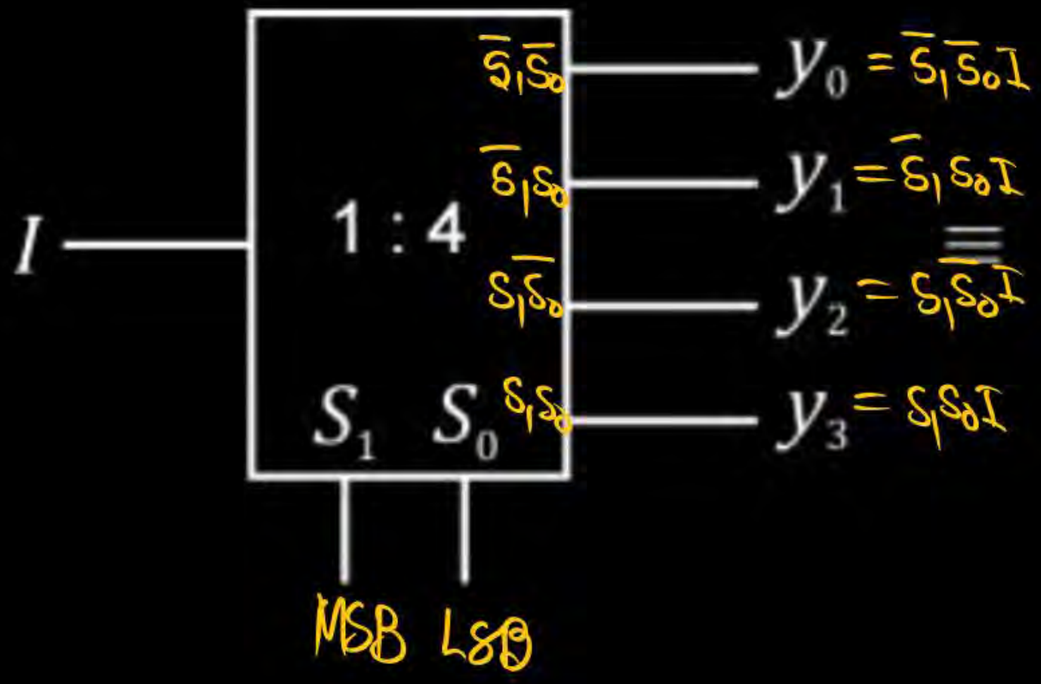
Step 5:

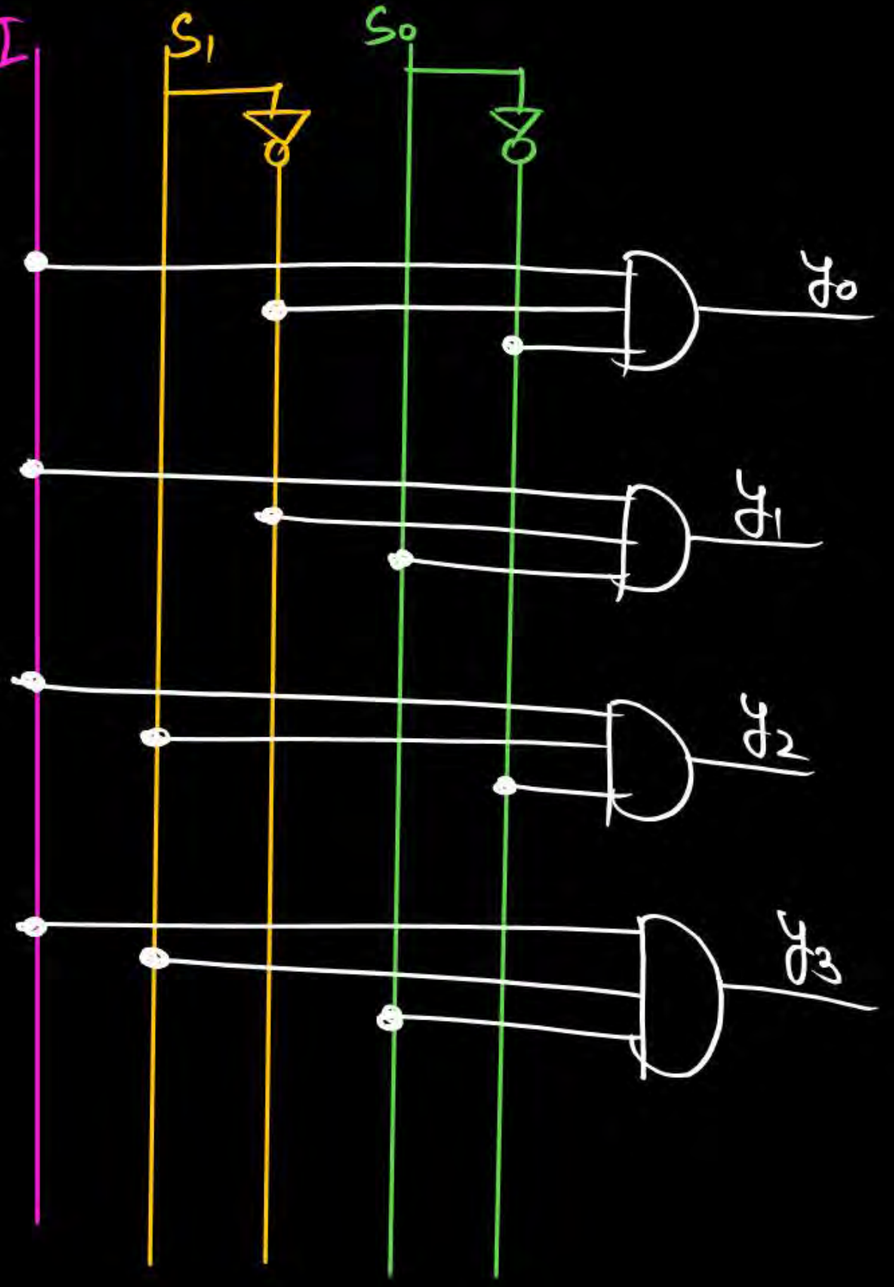


DEMUX

Design a 1 x 4 DE-MUX?

S_1S_0	y_0	y_1	y_2	y_3
0 0	I	0	0	0
0 1	0	I	0	0
1 0	0	0	I	0
1 1	0	0	0	I



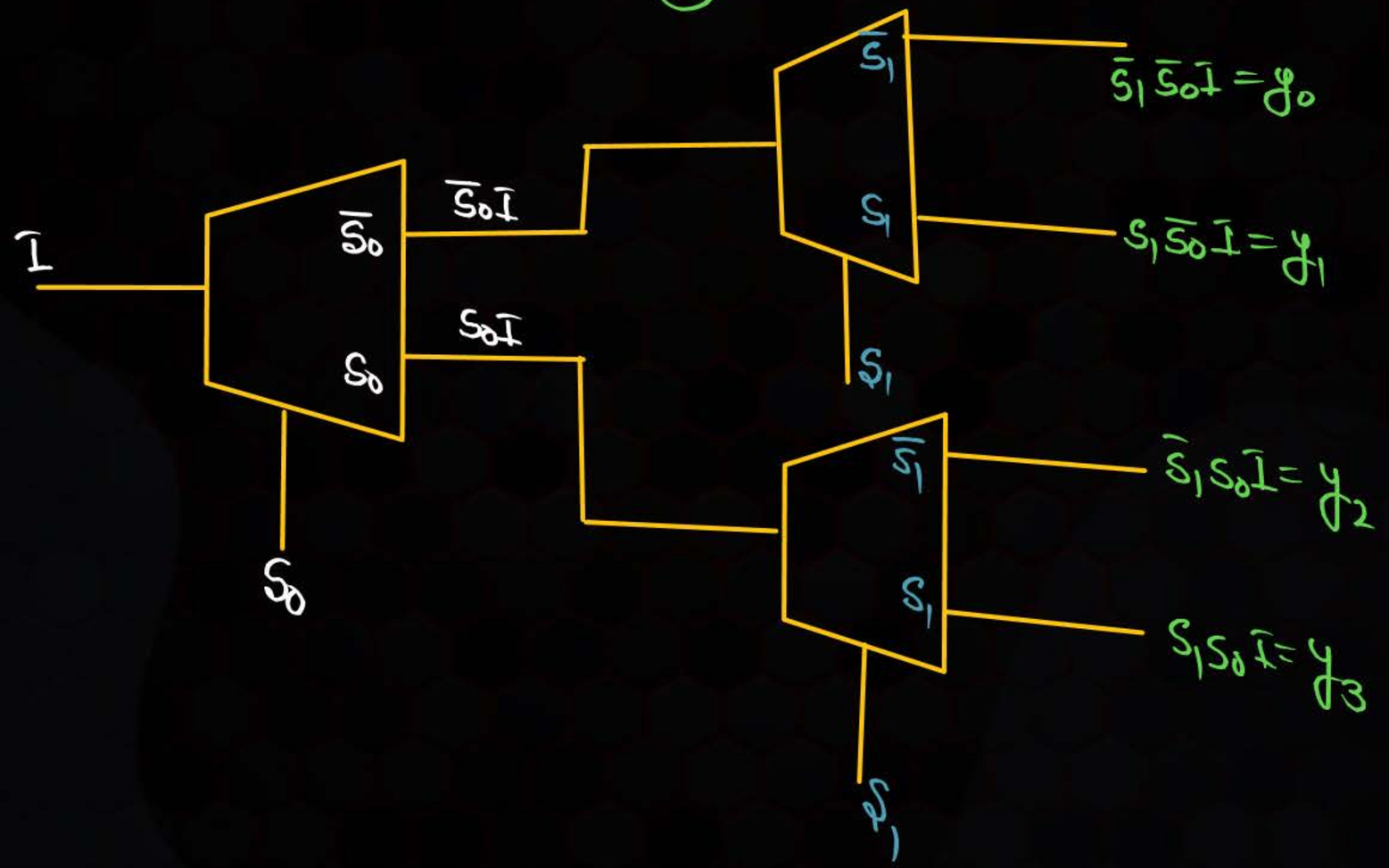


HW

- Q 1 x 8 DE-MUX
- Q 1 x 16 DE-MUX

1X2 DE-MUX $\xrightarrow{\frac{1}{2}+\frac{2}{2}}$ 1X4 DE-MUX

2+1 = 3



DEMUX

Higher order DeMUX using lower order DeMUX

- $1:4$ $\xrightarrow{\text{Using } 1:2}$ $\frac{4}{2}+\frac{2}{2}=3$
- $1:8$ $\xrightarrow{\text{Using } 1:2}$ $\frac{8}{2}+\frac{4}{2}+\frac{2}{2}=7$
- $1:16$ $\xrightarrow{\text{Using } 1:2}$ 15
- $1:2^n$ $\xrightarrow{\text{Using } 1:2}$ 2^n-1

	Base (Radix)	Digit
Binary	2	0, 1
	3	0, 1, 2
Quad	4	0, 1, 2, 3
	5	0, 1, 2, 3, 4
	6	0, 1, 2, 3, 4, 5
	7	0, 1, 2, 3, 4, 5, 6
Octal	8	0, 1, 2, 3, 4, 5, 6, 7
	9	0, 1, 2, 3, 4, 5, 6, 7, 8
	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9

$(101)_{2,3,4,5,\cdots}$

$(246)_{7,8,9,10,11\cdots}$

Base (Radix)	Digit
11	0,1,2,3,4,5,6,7,8,9, A^{10}
12	0 – 9, A, B
13	0 – 9, A, B, C
14	0 – 9, A, B, C, D
15	0 – 9, A, B, C, D, E
16	0 – 9, A, B, C, D, E, F

$(12C)$

13, 14, 15, ...

minimum Base

$(BF6)$

B



Base	$(r-1)$'s complement	r's complement
Base $(r) = 2$	1's	2's
Base $(r) = 8$	7's	8's
Base $(r) = 10$	9's	10's
Base $(r) = 16$	E's (15's)	F's (16's)

Base (r) = 2

1's complement

$$\begin{array}{r} {\scriptstyle 1} \\ -\ 0 \\ \hline 1 \end{array} \qquad \begin{array}{r} {\scriptstyle 1} \\ -\ 1 \\ \hline 0 \end{array}$$

Ex $101 \xrightarrow{1's} 010$

$$\begin{array}{r} 1\ 1\ 1 \\ -\ 1\ 0\ 1 \\ \hline 0\ 1\ 0 \end{array}$$

Base (r) = 8

7's complement

Ex $\equiv$ $373 \xrightarrow{\text{7's complement}} 404$

$$\begin{array}{r} 777 \\ -373 \\ \hline 404 \end{array}$$

Base (r) = 10 :→

9's complement

$$\begin{array}{r} 9 \\ -\,0 \\ \hline 9 \end{array} \quad \begin{array}{r} 9 \\ -\,9 \\ \hline 0 \end{array}$$

9 ⟷ 0

→ self complement

$$\begin{array}{r} 9 \\ -\,1 \\ \hline 8 \end{array} \quad \begin{array}{r} 9 \\ -\,8 \\ \hline 1 \end{array}$$

8 ⟷ 1

→ self complement

$$\begin{array}{r} 9 \\ -\,2 \\ \hline 7 \end{array} \quad \begin{array}{r} 9 \\ -\,7 \\ \hline 2 \end{array}$$

7 ⟷ 2

→ self complement

Decimal system

10^3 10^2 10^1 10^0 ← weight

$(2\ 3\ 6\ 9)_{10}$ ← coefficient

- weighted code
- self complemented

8 4 2 1

$(1\ 0\ 1\ 0\ 1)_2 = (21)_{10}$

ENCODER

ENCODER

- A circuit used to convert any code into Binary are called Encoder.
- 4 × 2 Encoder [Quad to Binary Encoder]
- 8 × 3 Encoder [Octa to Binary Encoder]
- 16 × 4 Encoder [Hexa to Binary Encoder]

8421

Decimal	BCD	EXCESS-3	4221
0	0000	0011	0000
1	0001	0100	0001
2	0010	0101	0010
3	0011	0110	0011
4	0100	0111	0110
5	0101	1000	1001
6	0110	1001	1100
7	0111	1010	1101
8	1000	1011	1110
9	1001	1100	1111

BCD → Binary coded Decimal

↳ Each decimal numbers are represented by 4 bits

0 → 0000
1 → 0001
2 → 0010
3 → 0011
4 → 0100

7 → 0111
8 → 1000
9 → 1001





ENCODER

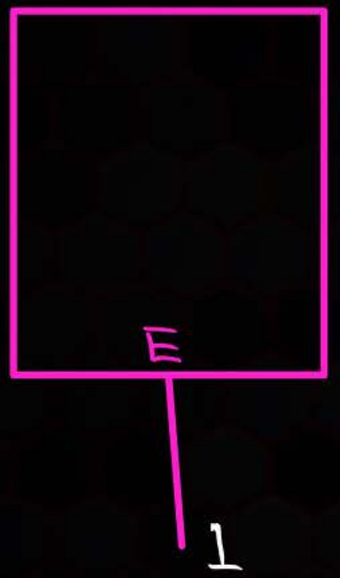
CODE CONVERSION

BCD CODE - It is weighted code but not self complimented

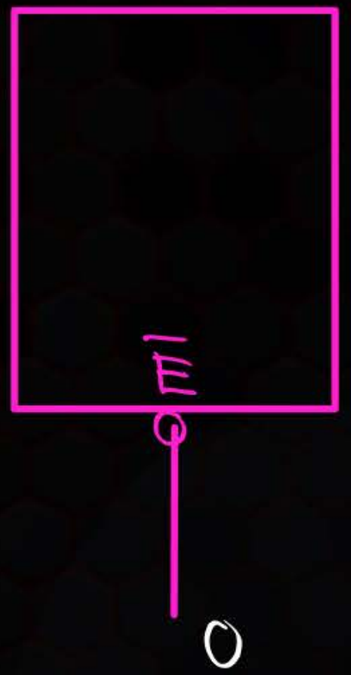
Excess 3 CODE - It is not a weighted code but self complimented code

4221 CODE - It is weighted code and self complimented code

Active high



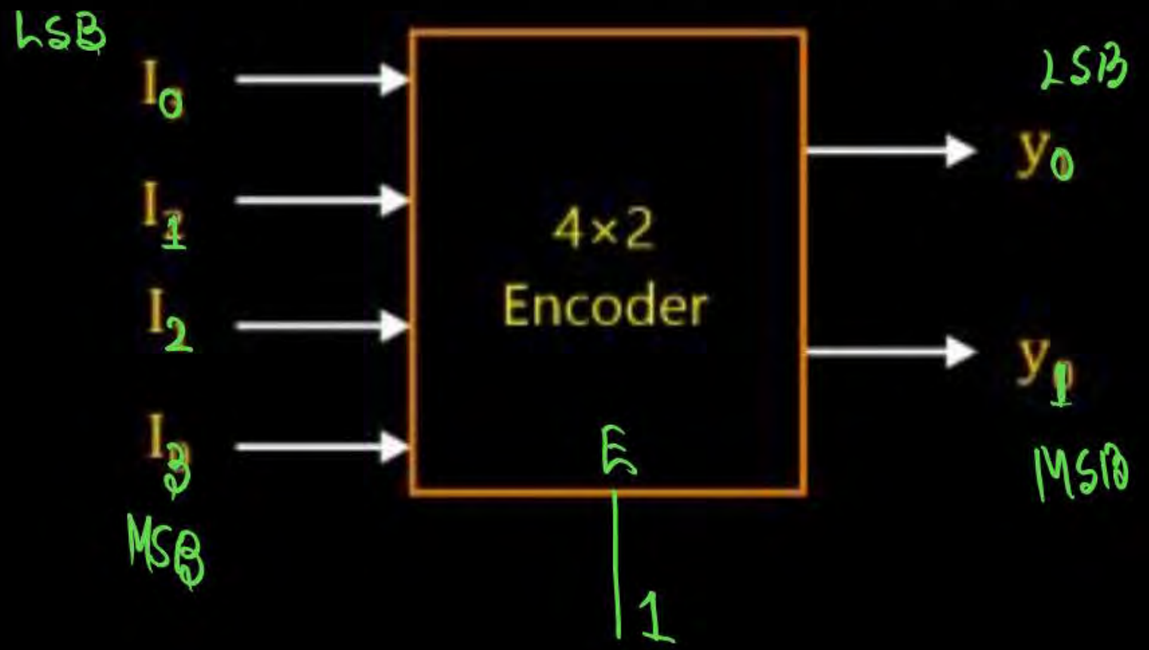
Active low



ENCODER

4 × 2 ENCODER

Step 1. Find the number of inputs and outputs



ENCODER

4 × 2 ENCODER

Step 2. Truth Table



MSB LSB

I_3	I_2	I_1	I_0	Y_1	y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

ENCODER

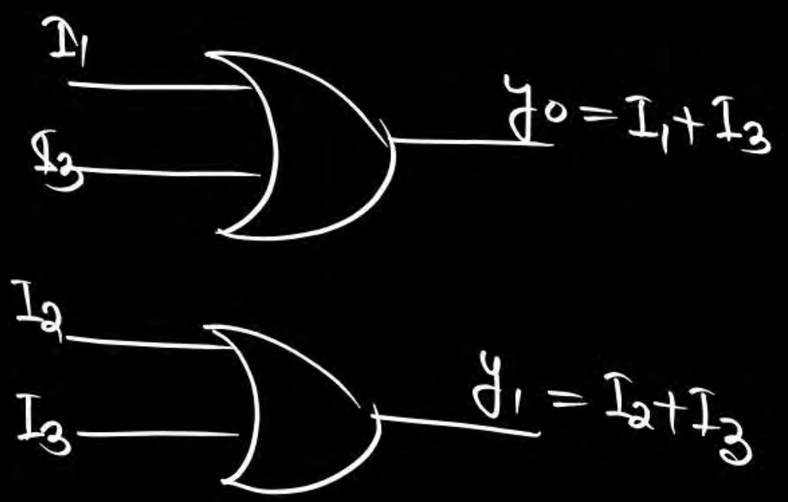
4 × 2 ENCODER

Step 3. $y_0 = I_1 + I_3$

$y_1 = I_2 + I_3$

Step 4: No minimization

Step 5:



ENCODER

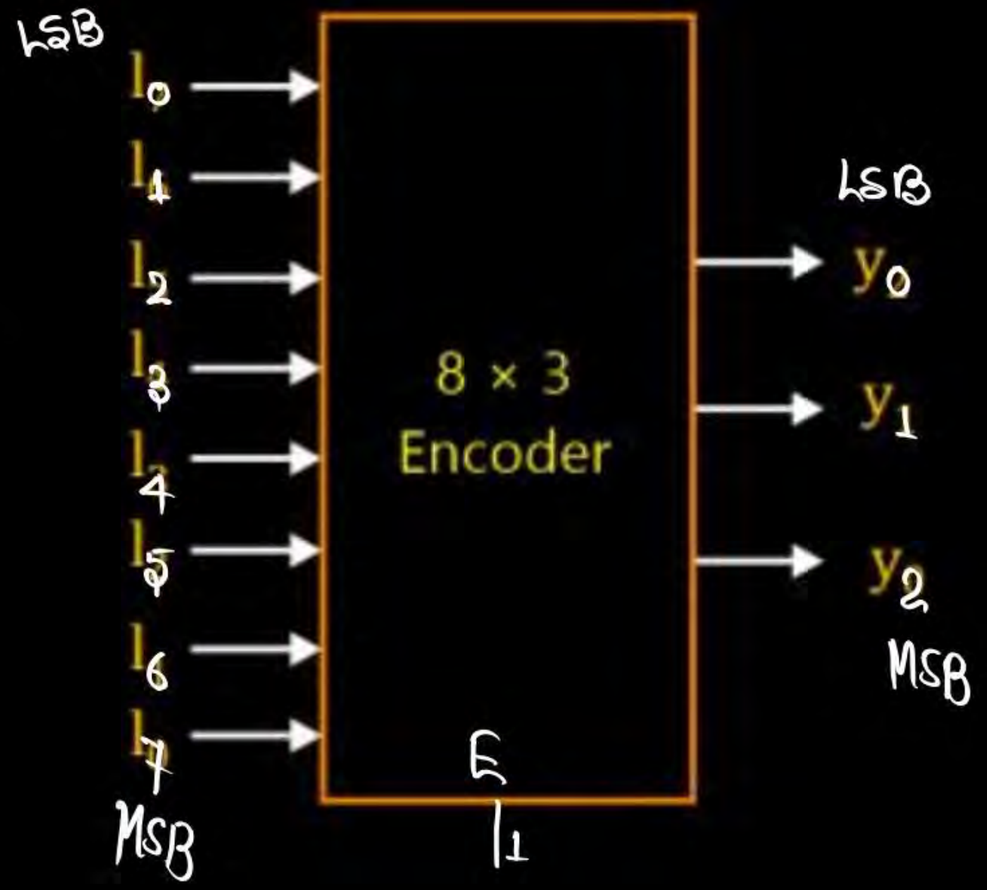
{ octal to Binary Encoder }

8 × 3 ENCODER

inputs

output

Step 1.



ENCODER

8 × 3 ENCODER

Step 2.

MSB LSB

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	y_2	y_1	y_0
1	0	0	0	0	0	0	0	1	1	1
0	1	0	0	0	0	0	0	1	1	0
0	0	1	0	0	0	0	0	1	0	1
0	0	0	1	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0

ENCODER

8 × 3 ENCODER

Step 3.

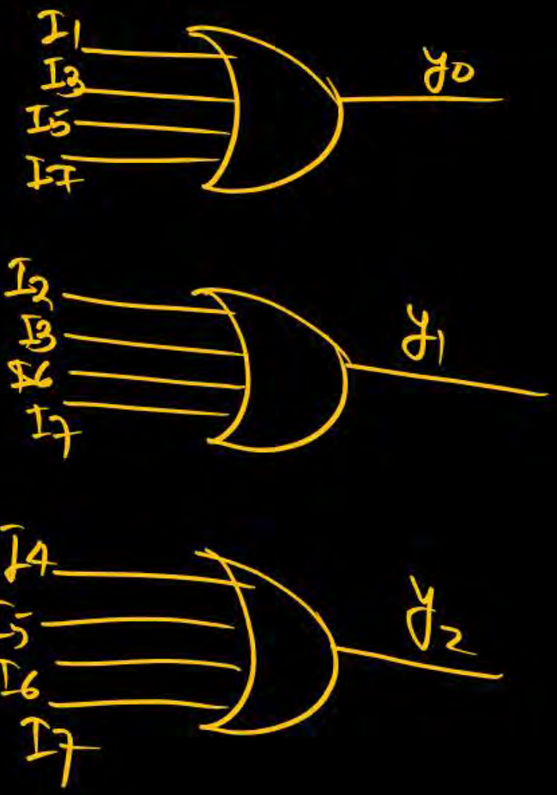
$y_0 = \bar{I}_1 + \bar{I}_3 + \bar{I}_5 + \bar{I}_7$

$y_1 = \bar{I}_2 + \bar{I}_3 + \bar{I}_6 + \bar{I}_7$

$y_2 = \bar{I}_4 + \bar{I}_5 + \bar{I}_6 + \bar{I}_7$

step4

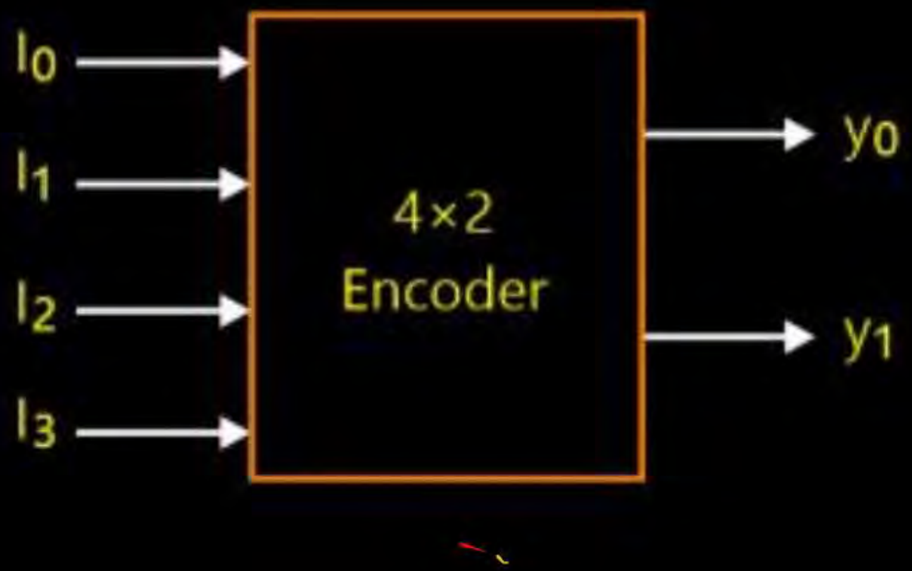
step 5:



Q Design 16 x 4 Encoder.

ENCODER

PRIORITY ENCODER

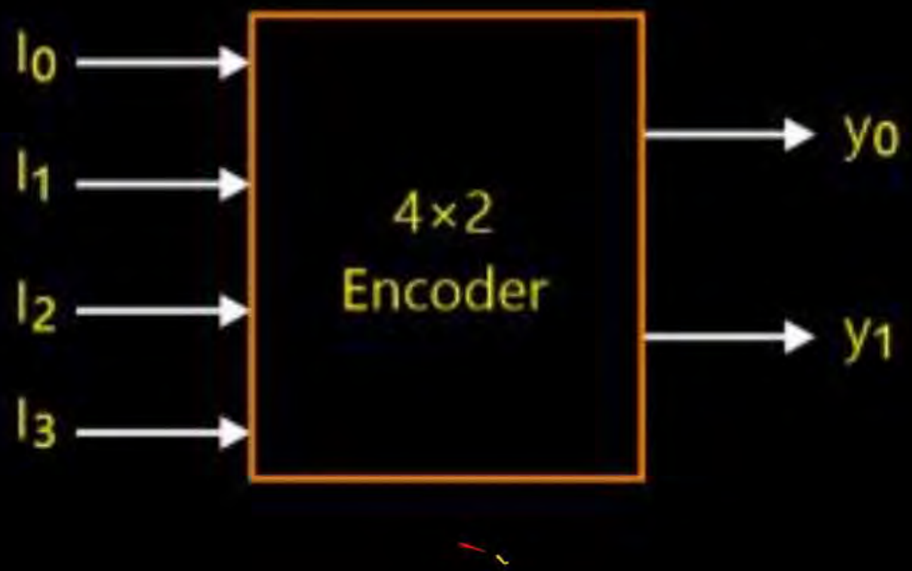


LSB priority

MSB			LSB	MSB	LSB
I_3	I_2	I_1	I_0	Y_1	y_0
X	X	X	1	0	0
X	X	1	0	0	1
X	1	0	0	1	0
1	0	0	0	1	1

ENCODER

PRIORITY ENCODER



MSB priority

MSB			LSB	MSB	LSB
I_3	I_2	I_1	I_0	Y_1	y_0
0	0	0	1	0	0
0	0	1	X	0	1
0	1	X	X	1	0
1	X	X	X	1	1

Thank you DOSTO